

Assessment #3 - Annotated Bibliographies

Student Name: Robin Kim

ISM I

Date: 8/31/26

Subject: Annotated Bibliographies #3

Reference 1

Scarborough, D. M., Colón, P. E., Linderman, S. E., & Berkson, E. M. (2021). Comparison of kinematic sequences during curveball and fastball baseball pitches. *Frontiers in Sports and Active Living*, 3, Article 699251. <https://doi.org/10.3389/fspor.2021.699251>

Using 3D motion-capture technology on 14 pitchers throwing both curveballs and fastballs, the study finds that nearly half of all pitches had a modified "kinetic sequence" associated with increased elbow and shoulder strain. The orthopedic surgeons from Massachusetts General Hospital and Harvard Medical School who conducted the research have no financial conflicts of interest, however the paper fails to address sample size implications. Only 4 out of 14 pitchers were high schoolers, and the paper does not separate results by groups, making it difficult to assess the risk specific to HS pitchers. That said, I am going to use the research to argue that the injury model with MLB players observations could fail to account for risk factors critical to HS pitchers.

Reference 2

Mair, H., Kenney, N. A., Uhl, T. L., Ullery, L. R., & Hosey, R. G. (2022). Effect of pitching restrictions and mound distance on youth baseball pitch counts. *Orthopaedic Journal of Sports Medicine*, 10(7), Article 23259671221110547. <https://doi.org/10.1177/23259671221110547>

A study comparing three youth baseball leagues (some with limits on pitches, some on innings, some with different mound distances) found that innings-limited ones allowed pitchers to throw more per outing than their pitch limited groups. The study was done by sports medicine doctors and athletic trainers at the University of Kentucky Medical Center, but one of them has to disclose a conflict of interest (funding). The comparison is between three leagues in the same city (may not be applicable elsewhere), but is nevertheless an interesting point that simply copying a professional rule is insufficient, since the two pitch-limited leagues had very different rules in this regard, which is exactly the point I am trying to make.

Reference 3

Kriz, P. K., Collins, C. L., Staffa, S. J., Zurakowski, D., Zaremski, J. L., Bowers, R. L., Kriz, J. P., & Sugimoto, D. (2026). Do pitching restriction policies reduce shoulder and elbow injuries in high school baseball players? *Orthopaedic Journal of Sports Medicine*, 14(2), Article 23259671251412404. <https://doi.org/10.1177/23259671251412404>

A 2016 NFHS policy, informed by MLB Pitch Smart recommendations, appeared to have few effects on the rate of shoulder and elbow injuries nationwide, despite specific elements being associated with decreased injury incidence. The lead author is a sports medicine physician at the Boston Children's Hospital, and the team reports no conflicts of

interest. An analysis using a large national injury database implies high generalizability and credibility. However, it is only able to demonstrate an association rather than cause-and-effect relationship because of the nature of correlational study. This finding will serve as an example where an MLB metric failed to capture the grand scheme of things but succeeded when sliced into individual HS components.

Reference 4

Matsuura, T., Takata, Y., Iwame, T., Iwase, J., Yokoyama, K., Takao, S., Nishio, S., Arisawa, K., & Sairyo, K. (2021). Limiting the pitch count in youth baseball pitchers decreases elbow pain. *Orthopaedic Journal of Sports Medicine*, 9(3), Article 2325967121989108.
<https://doi.org/10.1177/2325967121989108>

In 2018, one Japanese region replaced its innings-based pitch limit with a pitch-count-based limit; this paper follows the changes. Youth pitchers were able to throw less pitches on average and reported significantly less elbow pain. The authors, orthopedic surgeons at Tokushima University, found their results to be statistically significant with a very large sample size, and the paper explicitly cites no conflicts of interest. This conclusion comes with the caveat that the pitchers' studies were 8-12 years old, making it less applicable to HS pitchers. I will use this source to support that previous claim made in reference 3, as both demonstrate that MLB informed pitch limits do not always fit the needs of a specific league or age; adjustments to rules must be made at the regional and age levels.

Reference 5

Thomay, C. D. (2014). *Markov chain theory with applications to baseball* [Independent study thesis, College of Wooster]. OpenWorks.

<https://openworks.wooster.edu/independentstudy/5722/>

Unlike the journal articles described here, this source is an undergraduate mathematics thesis reviewed by a faculty advisor affiliated to the league it researches. It follows the same sabermetric run-expectancy calculation as the others, on 2014 College of Wooster games, finding that lower expected run gains from bunts and steals due to the team's own lower success rates. Its limited scope as one team's one season makes its result less significant, but, as a mathematical formula, its results would be no less certain to occur in any other team's games at any point in the future, making the mathematical findings still highly relevant after 12 years. I will use it to show that such sabermetric calculations are used across levels of play, when connected with the right data.

Research Article Links

1. <https://kami.app/b1K-pGJ-1fK-FLN>
2. <https://kami.app/9KG-8Vn-GB6-sQm>
3. <https://kami.app/vtY-WRZ-p7E-sg8>
4. <https://kami.app/R54-1ZB-dFt-YTd>
5. <https://kami.app/Ysz-PGz-vBK-1hz>